

Geometry Unit 13 Assessment Guide

General Information	Unit 13 is a comprehensive unit that revisits many concepts students have learned throughout the course. The benchmarks MA.912.GR.3.2, MA.912.GR.3.3, and MA.912.GR.3.4 require a wide range of knowledge and understanding from students. A unit assessment cannot cover every possible type of question that aligns with these benchmarks. Therefore, the unit assessment contains questions that educators can use to help evaluate student understanding of coordinate geometry while also considering the length of the assessment. Teachers may wish to provide grid paper for students to use during the assessment. Teachers are encouraged to modify the assessment or to use formative assessments that may further evaluate a student's understanding of using coordinate geometry. As the teacher looks at student work, some of the steps students have to apply are the same steps students would have to apply for most of the questions for that focus of that particular benchmark. For example, if students are given four vertices and are asked to determine the type of quadrilateral, then the students should find the slopes and distances of the sides and/or the diagonals to help determine the type of quadrilateral. Teachers can use student work to determine if students are struggling with the mathematics of finding the slope or the distance or if the student can find these values but struggle using that information to define the quadrilateral using properties or theorems. Students who struggle with the properties or theorems may benefit the most from reteaching or a review activity. Such efforts may help students prepare for the End-of-Course assessment.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.912.GR.3.1, MA.912.GR.3.3
	$(-6, 4)$	Recommended Scoring (18 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider looking for the following: - Shows knowledge that $ND:NW$ is 1:3. (3 points) - Finds distance between x values. (3 points) - Finds distance between y values. (3 points) - Applies correct direction from N to find the x-coordinate. (3 points) - Applies correct direction from N to find the y-coordinate. (3 points) - Correct answer (3 points)
Question	2	Benchmark(s)	MA.912.GR.3.2, MA.912.GR.3.3, MA.912.GR.7.2
	☐ $(-5, 0)$ ☒ $(-4, -5)$ ☐ $(-2, -3)$ ☒ $(-2, -5)$ ☒ $(0, -1)$ ☐ $(8, 7)$	Recommended Scoring (21 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider looking for the following: - Correctly writes the equation $(x + 3)^2 + (y + 2)^2 = 10$. (3 points) - Determines whether or not each point is on the circle using the equation. (3 points each)

Question	3	Benchmark(s)	MA.912.GR.3.2, MA.912.GR.3.3
	☐ kite ☐ square ☐ rhombus ☐ rectangle ☐ trapezoid ☐ parallelogram	Recommended Scoring (27 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider looking for the following: • A sketch of the quadrilateral. • The slope of each side. Alternatively, the slopes of the diagonals. (12 points) • The length of each side. Alternatively, the lengths of the diagonals. (12 points) • Correct Answer (3 points)
Question	4a	Benchmark(s)	MA.912.GR.3.2, MA.912.GR.3.3
	☐ scalene ☐ isosceles ☐ equilateral ☐ right ☐ acute ☐ obtuse	Recommended Scoring (22 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider looking for the following: • A sketch of the triangle. • The slope of each side. (9 points) • The length of each side. (9 points) • Correct Answer (2 points each)

Question	4b	Benchmark(s)	MA.912.GR.3.2, MA.912.GR.3.3
	$\left(-1\frac{2}{3}, 0\right) \text{ or } \left(-\frac{5}{3}, 0\right)$	Recommended Scoring (3 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider looking for the following: • The sum of the x-coordinates. • The sum of the y-coordinates. • Correct Answer Note: Students may use different methods to find the centroid.
Question	5	Benchmark(s)	MA.912.GR.3.4
	1312.5 mi^2	Recommended Scoring (9 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider looking for the following: • The area of the rectangle. • The area of the trapezoid. • Correct Answer Students may make a sketch and may find the area differently.